TRINITY INTERNATIONAL COLLEGE

Dillibazar Height, Kathmandu

Final Assessment 2025

BCA (Batch-2080)

Semester: **Fourth**
Subject: **Numerical Methods**
Course No: **CACS 252**

Full Marks: **50**
Pass Marks: **24**
Time: **2:50 Hrs**

Candidates are required to give answers in their own words as far as practicable. The figures in the margin indicate full marks.

Spying capturing model real world data
Design analysis of system

Group 'B'

Attempt Any Six Questions. **[5x6=30]**

1. What are the importance of Numerical Method in the field of science and engineering? Define error in Numerical method.
2. What are the limitations of newton -Raphson method? Using this method, the find the real root of the equation cosx-3x+1=0, correct to four decimal places.

   optimization 2 decision making
   accuracy & flexibility

3. Solve using partial pivoting method

   or $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

   2x+y+z-2u=-10
   4x+2z+u=8
   3x+2y+2z=7
   x+3y+2z-u=-5
4. Estimate the value of y(4.5) using Newton difference interpolation formula.

X	1	2	3	4	5
Y	0.5	2	4.5	8	12.5

Require Derivative
May not converge
Fails when denominator is zero or near to
Computation cost

5. Evaluate $\int_{-1}^{2} e^{-x^2} dx$ using gaussian 3-point formula.
6. Derive the general Newton-Cote quadrature formula and use it obtain Simpson's 1/3 formula.
7. Write a pseudo-code to solve first order differential equation using RK-4 method.

Group 'C'

Attempt Any Two Questions. **[10x2=20]**

8. Estimate y(4) using cubic spline interpolation technique from the following data.

X	3	5	7	9	11
Y	6	9	12	9	6

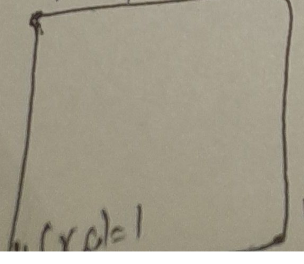
9. *Write the differences between initial value problem and boundary value problem.Solve the following boundary value problem using shooting method by dividing the interval into four sub intervals.*

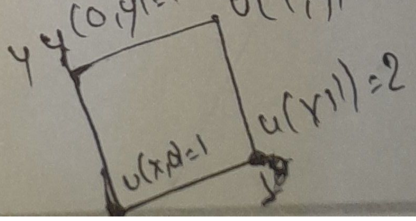
   Y''+3Y'-Y=Cosx   y(0)=2 and y(2)=3

10. *Derive the Laplace equation.*

Solve the discretized form of Laplace's equation, ∂ 2 u ∂ x 2 ⊚ ∂ 2 u ∂ y 2 = 0 , for u(x,y) defined within the domain of 0 ≤ x ≤ 1 and 0 ≤ y ≤ 1, given the boundary conditions (I) u(x, 0) = 1 (II) u (x,1) = 2 (III) u(0,y) = 1 (IV) u(1,y) = 2

Good Luck





TRINITY INTERNATIONAL COLLEGE

Dillibazar Height, Kathmandu

Final Assessment 2025

BCA (Batch-2080)

TRINITY INTERNATIONAL COLLEGE

Semester: **Fourth**

Subject: **Database Management System**

Course No: **CACS 255**

SET - A

Full Marks: **60**

Pass Marks: **24**

Time: **3 Hrs.**

Candidates are required to give answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group 'B'

Attempt any SIX questions. [6×5=30]

2. Explain the advantages of DBMS over traditional file system.
3. Explain the client-server architecture in DBMS with its advantages.
4. Why do we need query optimization? Explain cost-based optimization. → Heuristic
5. Define Data Abstraction. Explain logical and physical data independence. [2+3]
6. How candidate key is different from primary key? Also, explain foreign key in details. [1+4]
7. Describe the concept of serializability in transaction processing. [2+3]
8. Explain the role of DBA in detail. → performance tuning → Troubleshoot → Security → Coordination b/w → Integrity → Backup & recovery → Data integrity

Group C

Attempt any TWO questions. [2×10=20]

9. Design an ER diagram for Hospital System. Use your assumption for the selection attributes, entities and relationships. Show the use of total and partial participation along with the appropriate cardinalities.

10. Convert following table to 3NF.

Student_ID	Student_Name	Prof_ID	Prof_Name	Grade
1	John	101	Dr. Shrestha	A
2	Ram	102	Dr. Sharma	A
3	Hari	103	Dr. Bhusal	C
4	Krishna	104	Dr. Rimal	D
5	Tom	105	Dr. Malla	B

11. Given the Schema as below, write relational algebra and SQL queries for following:

Sailors (<u>Sid</u>, Sname, rating, age)

Boats (<u>bid</u>, bname, colour)

Reserves (<u>Sid</u>, <u>bid</u>, reservedate)

a) Find name, rating and age of all the sailors.

b) Find the name and color of boats having color blue.

c) Find name of the boats reserved on date 2025/07/20.

d) Find the names of boats and sailors who have reserved boats having color green

e) Use Left and Right Outer Join between sailors and Reserves.

GOOD LUCK

TRINITY INTERNATIONAL COLLEGE

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Final Assessment 2025

BCA (Batch-2080)

SET A

Semester: **Fourth**

Subject: **Operating System**

Course No: **CACS 251**

Full Marks: **60**

Pass Marks: **24**

Time: **3 Hrs**

Candidates are required to give answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group 'B'

Attempt Any Two Questions. **[10x2=20]**

1. Discuss about single level and two level directory system. Consider the following process and answer the following questions.

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				
P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

a. What is the content of matrix Need?

b. Is the system in safe state?

c. If P1 request (0,4,2,0) can the request be granted immediately.

2. When does race condition occur in inter process communication? What does busy waiting mean and how it can be handled using sleep and wakeup strategy?
3. Define shell and system call. suppose a disk has 201 cylinders, numbered from 0 to 200. At same time the disk arm is at cylinder 95, and there is a queue of disk access requests for cylinders 82,170,43,140,24,16 and 190. Calculate the seek time for the disk scheduling algorithm FCFS,SSTF,SCAN and C-SCAN.

Group 'C'

Attempt Any Six Questions. **[5x6=30]**

4. Distinguish between starvation and deadlock . How does the system schedule process using multiple queues?
5. List any two demerits of disabling interrupt to achieve mutual exclusion. Describe about fixed andvariable partitioning + dynamic hole
6. For the following dataset, compute average waiting time for SRTN and SJF. 7

Process	Arrival Time	Burst Time preemp
P0	0	7
P1	2	4
P2	4	1
P3	5	4

7. Discuss the advantages disadvantages of implementing file system using Linked List.
8. Consider the page references 7,0,1,2,0,3,0,4,2,3,0,3,2, Find the number of page fault using OPR and FIFO, with 4 page frame.
9. Describe the working mechanism of DMA.
10. What is the task of disk controller ? List some drawback of segmentation.
11. Write the structure and advantages of TLB.
12. Why do we need the concept of locality of reference ? List the advantages and disadvantages of Round Robin algorithm.

Good Luck

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Final Assessment 2025

BCA (Batch-2080)

SET A

TRINITY INTERNATIONAL COLLEGE

Semester: **Fourth**

Subject: **Software Engineering**

Course No: **CACS 253**

Full Marks: **50**

Pass Marks: **24**

Time: **2:50 Hrs**

Candidates are required to give answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group 'B'

Short Answer Questions.

Attempt Any Six Questions.

[6x5=30]

1. What is extreme programming? Explain how the principle underlying extreme programming lead to the accelerated development of software.
2. Differentiate between verification and validation. Explain why software inspection is an effective technique for discovering errors in software?
3. What is architecture design? Explain layered model of software architecture with example.
4. Discuss COCOMO model in cost estimation of the software in detail.
5. What is CASE? Explain the importance of CASE tools in software development life cycle.
6. What is software engineering? Explain the challenges facing software engineering.
7. Define quality assurance? Under what circumstances would you recommend the use of staged representation of the CMM?

Group'C'

Long Answer Questions.

Attempt Any Two Questions. [2x10=20]

8. What is Capability Maturity Model? Explain its levels along with key process area. Differentiate between alpha and beta testing. [2+4+4]
9. What do you mean by software requirement? Differentiate between functional requirement and non-functional requirement. List the functional requirement of online examination system and represent it in use case diagram.[2+4+4]
10. What do you mean by software maintenance? Explain the different types of maintenance? Explain reengineering process in detail. [2+4+4]

Good Luck

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Final Assessment 2025

BCA (Batch-2080)

SET A

Semester: **Fourth**		Full Marks: **50**
Subject: **Scripting Language**		Pass Marks: **24**
Course No: **CACS 254**		Time: **2:50 Hrs**

Candidates are required to give answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group 'B'

Short Answer Questions **[6x5=30]**

1. Why javascript is called a loosely typed language? What are the potential platforms where JavaScript can be used? (5)
2. Explain different ways to add javascript in an html file with example of each. (5)
3. Create an html form to accept user values from the user with proper validation using javascript. Validation should be as follows (2+1+2)
   a) All the field is required
   b) length of password should be a minimum 8 characters.
   c) on submit display information in alert box if validation is passed Explain MySQL Joins with an example query.
4. Define AJAX. How does it improve the performance of a web application?
5. Write the syntax and use of switch in JavaScript with an example.
6. Explain Exception Handling in PHP. Provide an example using try, catch, and finally.
7. Define and explain the concept of encapsulation in PHP. Provide an example code snippet to illustrate how encapsulation is implemented in a PHP class.

Group 'C'

user interface elements

Long Answer Questions **[2x10=20]**

8. Explain the **Document Object Model (DOM)** in JavaScript. Discuss its role in event handling with examples.
9. Explain the concept of inheritance in PHP with proper example code and write its advantages in object-oriented programming.
10. Write down the PHP script to display data from a database table to a web page in the following format. Make your own assumption about the database connectivity.

Id	Product Name	Category	Brand	Quantity

Switch
encap
inher
Excep